

SMART INDIA HACKATHON 2026

Internal Hackathon Idea Submission · Problem Statement 26102

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MPLADS-SATARK (सतर्क)

AI Risk Scoring, Multi-Layer Anomaly Detection & Proactive Audit Engine

Problem Statement ID	26102
Problem Statement Title	Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation
Ministry / Organization	Ministry of Statistics & Programme Implementation (MoSPI / DIID)
Theme & Category	Clean & Transparent Governance / Smart Automation · Category: Software
Team Name & Team Lead	[Insert Registered Team Name] · Lead: Purva Shewale (Full-Stack & System Architect)
Team Members	Vaibhav (Frontend Lead), Tanvi (AI/ML), Ankesh (Graph Networks), Makarand (Forensics), Raj (Statutory)

PROPOSED SOLUTION

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Idea Overview, Root-Cause Problem Resolution & Key Innovations

1. SOLUTION DESCRIPTION

- **Real-Time Vigilance Platform:** Replaces slow historical ledgers with automated fraud prevention *before* funds are disbursed.
- **6 Independent Signals:** Evaluates every project proposal across pricing, networks, guidelines, and mathematics.
- **Explain-My-Score Waterfall:** Scores 0 to 100 with itemized additive points citing exact GFR 2017 clauses.

2. PROBLEMS ADDRESSED

- **Double-Invoicing:** Compares road/hall works with ISRO Bhuvan & MGNREGA to catch cross-scheme billing.
- **Inflated Estimates:** Benchmarks costs against CPWD Delhi Schedule of Rates (DSR) to flag overpricing.
- **Vendor Cartels:** Network graph uncovers single-agency fund monopolies (70%+ budget share).
- **Zombie Projects:** Survival hazard models predict stalled projects 6 months before deadlines lapse.

3. INNOVATION & UNIQUENESS

- **100% Real Government Data:** Validated on 176,925 works and 109,521 payment vouchers from live MoSPI endpoints.
- **Data Provenance Framework:** Every number carries an immutable badge (REAL, DERIVED, ESTIMATED, STATUTORY). Zero black boxes.
- **1-Click Inspection Dossier:** Pre-filled official memorandum for District Engineers to verify physical assets.

💡 **The Core Differentiator:** While other systems report what happened years ago, SATARK equips administrators with audit-defensible proof grounded in GFR 2017 rules before public money is lost.

TECHNICAL APPROACH

Architecture, 6-Layer Detection Signals & Working Prototype

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1. TECHNOLOGY STACK

- **Backend:** Node.js runtime, high-performance in-memory columnar store (sub-second query across 1.76L records), REST API.
- **Frontend:** Modern Vanilla CSS design system, Leaflet GIS geospatial map, Vis.js interactive network graphs, Chart.js.
- **ML & Mathematical Algorithms:** Tree-based Isolation Forest, Louvain Community Detection, Benford's Law (Chi-Square), Cox Proportional Hazards.

2. COLD PATH VS HOT PATH ARCHITECTURE

- **Cold Path (Batch Analytics):** Ingests nationwide MoSPI, CPWD DSR, and Bhuvan data offline into indexed cubes for instant dashboard rollups.
- **Hot Path (Live Demo < 15ms):** Evaluates newly injected project tenders live on stage in under 15ms (surpassing < 50ms SLA).
- **Focus & Context Ego-Graph:** Renders 1-2 hop neighborhoods (10-15 nodes) on demand to guarantee zero browser lag.

3. THE 6-SIGNAL MULTI-LAYER AI DEFENSE ENGINE

- SIG 1

Isolation Forest: CPWD cost overruns & delay days.
- SIG 2

Ego-Graph: Contractor cartel concentration & HHI.
- SIG 3

Statutory Rules: SC/ST quotas & Negative List.
- SIG 4

Dedup Engine: Cross-scheme Bhuvan match.
- SIG 5

Benford's Law: ₹4.98L split tender evasion.
- SIG 6

Survival Hazard: Zombie stalled works alert.

FEASIBILITY AND VIABILITY

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Scale Validation, Potential Risks & Strategic Mitigations

1. FEASIBILITY AT GOVERNMENT SCALE

- **Demonstrated Nationwide Scale:** Successfully indexed and analyzed 100% of India (36 States & UTs, 541 Constituencies, 176,925 works) in 1.8 seconds.
- **Zero Costly Infrastructure:** Operates on standard lightweight server infrastructure without expensive cloud GPU clusters or paid third-party APIs.
- **Seamless Adoption:** Connects directly to existing MoSPI eSAKSHI data pipelines without requiring workflow replacement.

2. CHALLENGES & STRATEGIC MITIGATIONS

- **Challenge 1: Scale Risk (27,000 Graph Nodes Freezing UI)**
Mitigation: Focus & Context pattern renders only 1-2 hop neighborhood (10-15 nodes) on demand instead of the full national graph.
- **Challenge 2: Changing Statutory Rules (GFR / MPLADS Revisions)**
Mitigation: Policy-as-Data architecture stores rules with effective date ranges; past flags remain audit-defensible years later.
- **Challenge 3: Cross-Department Name Discrepancies**
Mitigation: Hierarchical blocking (State + District block) followed by n-gram token similarity eliminates false matches.

⚙️ **Viability Guarantee:** Every architectural decision maps directly to a recognized failure mode in public sector software deployments: scale lag, guideline drift, and cross-department data noise.

IMPACT AND BENEFITS

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Stakeholder Transformation, Economic Savings & Social Justice

1. IMPACT ON 4 KEY STAKEHOLDERS

- **Union Ministry / DIID:** Real-time national macro visibility over ₹7,908 Cr in sanctioned funds, fund velocity tracking, and inter-state benchmarking.
- **State Nodal Authority:** Inter-district performance roll-up, agency concentration heatmaps, and unspent balance tracking.
- **District Magistrate / Collector:** Prioritized inspection dispatch queue reduces manual field verification workload by 85%.
- **Hon'ble Member of Parliament:** Self-check compliance scorecard with live progress meters for mandatory SC (15%) and ST (7.5%) quotas.

2. QUANTIFIABLE BENEFITS

- **Economic Impact:** Prevents an estimated ₹500+ Crore in annual public fund leakages (overpriced tenders, duplicate billing, and March rush dumping).
- **Social Justice:** Real-time statutory enforcement of mandatory allocations for Scheduled Caste (15%) and Scheduled Tribe (7.5%) communities.
- **Constitutional Realism:** Strict Human-in-the-Loop governance; AI assists officers with legal evidence but never freezes funds unilaterally.
- **Citizen Trust:** Radical transparency ensuring sanctioned roads, hospitals, and drinking water facilities physically exist.

 **Ground Reality:** When ₹500 Crore of leakages is prevented, it translates directly into thousands of rural drinking water systems, village roads, and classroom upgrades actually reaching citizens.

RESEARCH AND REFERENCES

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Primary Government Sources, Statutory Frameworks & Scientific Literature

1. PRIMARY GOVERNMENT DATA

- **MoSPI eSAKSHI Portal:** Live endpoints (<https://mplads.mospi.gov.in/>) for Works Recommended, Completed, and Expenditures.
- **CPWD Rates:** Delhi Schedule of Rates (DSR 2023-24) & State Cost Multipliers for all 36 States & UTs.
- **ISRO Bhuvan / MoRD:** Bhuvan Geoportal & MGNREGA Rural Asset Registry.

2. STATUTORY & FINANCIAL RULES

- **Ministry of Finance:** General Financial Rules (GFR 2017) — Rule 62 (Pacing Caps), Rule 99 (Vouching), Rules 130 & 139 (Works Execution).
- **MoSPI Guidelines:** Revised MPLADS Scheme Guidelines 2023 — Chapter 2 (SC/ST Quotas) & Annexure-I (Negative List).
- **CPWD Works Manual 2019:** Standard measurement book procedures.

3. SCIENTIFIC LITERATURE

- **Liu, Ting, Zhou (2008):** *Isolation Forest* (IEEE ICDM) — Multivariate tree-based anomaly isolation.
- **Newcomb (1881) & Benford (1938):** *The Law of Anomalous Numbers* — Forensic digit distribution $P(d) = \log_{10}(1 + 1/d)$.
- **Cox, D.R. (1972):** *Regression Models and Life-Tables* (JRSS) — Survival hazard modeling.

✓ **Audit-Defensible Standard:** Every detection rule, pricing benchmark, and fraud score in MPLADS-SATARK cites primary Government of India statutory documentation and verified scientific literature.